

11. Create a dataframe of ten rows, four columns with random values. Convert some values to nan values. Write a Pandas program which will highlight the nan values.

Aim

To create a Pandas DataFrame with random values, convert some values into NaN, and highlight the NaN values.

Procedure

1. Import the **pandas** and **numpy** libraries.
2. Create a DataFrame with **10 rows and 4 columns** using random values.
3. Convert selected values into **NaN**.
4. Define a function to identify **NaN** values.
5. Apply styling to highlight the **NaN** values.
6. Display the styled DataFrame.

```
exp11.py - C:/Users/sandr/exp11.py (3.12.10)
File Edit Format Run Options Window Help

import pandas as pd
import numpy as np

# Create DataFrame with 10 rows and 4 columns
df = pd.DataFrame(np.random.randn(10, 4),
                  columns=['A', 'B', 'C', 'D'])

# Convert some values to NaN
df.iloc[2, 1] = np.nan
df.iloc[5, 2] = np.nan
df.iloc[7, 0] = np.nan

# Highlight NaN values
def highlight_nan(val):
    if pd.isna(val):
        return 'background-color: yellow'
    return ''

print(df)

# Display highlighted DataFrame
df.style.map(highlight_nan)
```

```
IDLE Shell 3.12.10
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit
(AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/sandr/exp11.py =====

```

	A	B	C	D
0	-0.417149	-0.523296	-0.983372	0.988951
1	-0.538099	-0.145389	0.527635	0.719068
2	2.886760	NaN	-0.184112	-1.678215
3	1.214458	0.773820	0.077935	-0.226541
4	-0.764613	-0.661699	-0.022456	0.531621
5	-0.103656	-0.438136	NaN	-1.196735
6	1.824441	-0.104544	-0.696280	0.516115
7	NaN	0.939672	0.517156	0.941613
8	-0.440569	-0.796235	0.198123	-0.087037
9	-0.149489	-1.308687	-0.795491	-0.996277

Result

Thus, a DataFrame with **10 rows and 4 columns** was created successfully, and the **NaN** values were highlighted.

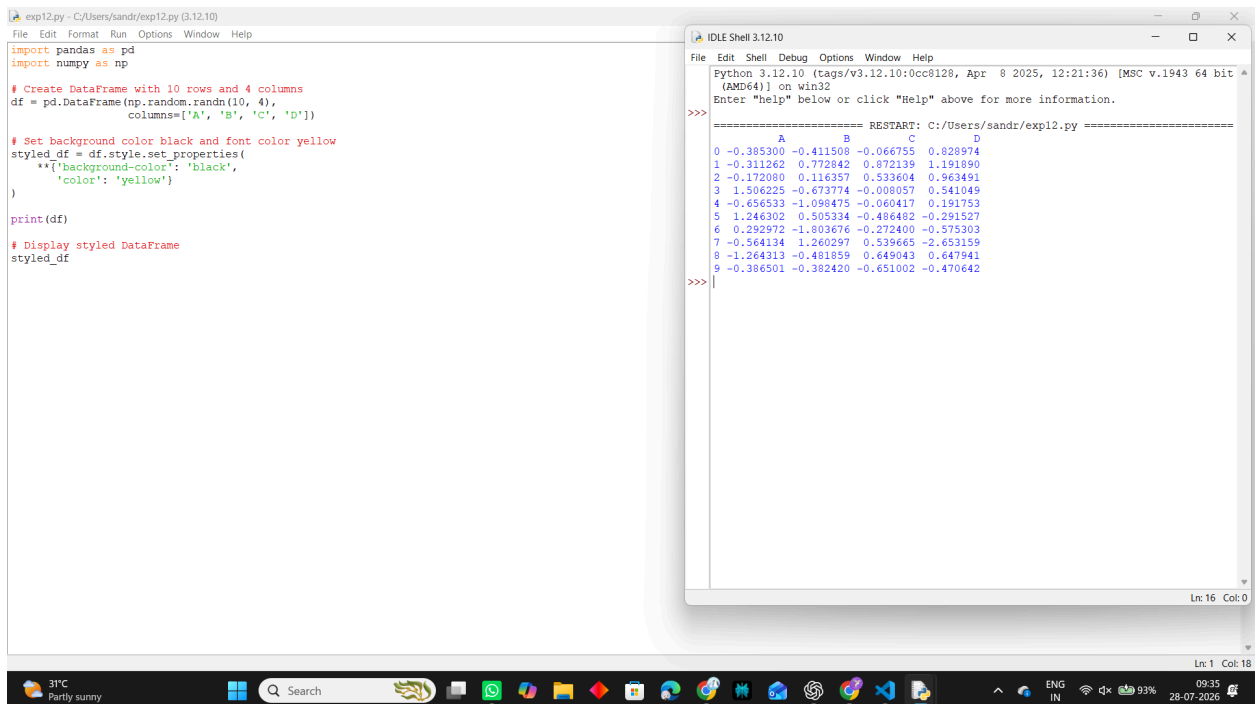
12. Create a dataframe of ten rows, four columns with random values. Write a Pandas program to set dataframe background Color black and font color yellow.

Aim

To create a Pandas DataFrame with random values and set its background color to **black** and font color to **yellow**.

Procedure

1. Import the **pandas** and **numpy** libraries.
2. Create a DataFrame with **10 rows and 4 columns** using random values.
3. Use the Pandas **style** property to customize the DataFrame.
4. Set the background color to **black**.
5. Set the font color to **yellow**.
6. Display the styled DataFrame.



```
exp12.py - C:/Users/sandr/exp12.py (3.12.10)
File Edit Format Run Options Window Help

import pandas as pd
import numpy as np

# Create DataFrame with 10 rows and 4 columns
df = pd.DataFrame(np.random.randn(10, 4),
                  columns=['A', 'B', 'C', 'D'])

# Set background color black and font color yellow
styled_df = df.style.set_properties(
    **['background-color': 'black',
       'color': 'yellow']
)

print(df)

# Display styled DataFrame
styled_df
```

```
IDLE Shell 3.12.10
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit
(AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/sandr/exp12.py =====
      A         B         C         D
0 -0.385300 -0.411508 -0.066755  0.828974
1 -0.311262  0.772842  0.872139  1.191890
2 -0.172080  0.116357  0.533604  0.963491
3  1.506225 -0.673774 -0.008057  0.541049
4 -0.656533 -1.098475 -0.060417  0.191753
5  1.246302  0.505334 -0.486482 -0.291527
6  0.292972 -1.803676 -0.272400 -0.575303
7 -0.564134  1.260297  0.539465 -2.653159
8 -1.264313 -0.481859  0.649043  0.647941
9 -0.386501 -0.382420 -0.651002 -0.470642

>>>
```

31°C Partly sunny 09:35 28-07-2026

Result

Thus, a DataFrame with **10 rows and 4 columns** was created successfully with a **black background and yellow font**.